

EXPERIMENT 32

Unusual Outbound Communication Detection

AIM

To detect outbound network records that differ significantly from a basic normal communication profile.

ALGORITHM

1. Define normal destination, protocol, and data-volume ranges.
2. Read simulated outbound records.
3. Compare each record with the baseline.
4. Flag records violating one or more characteristics.
5. Display the reasons for each alert.

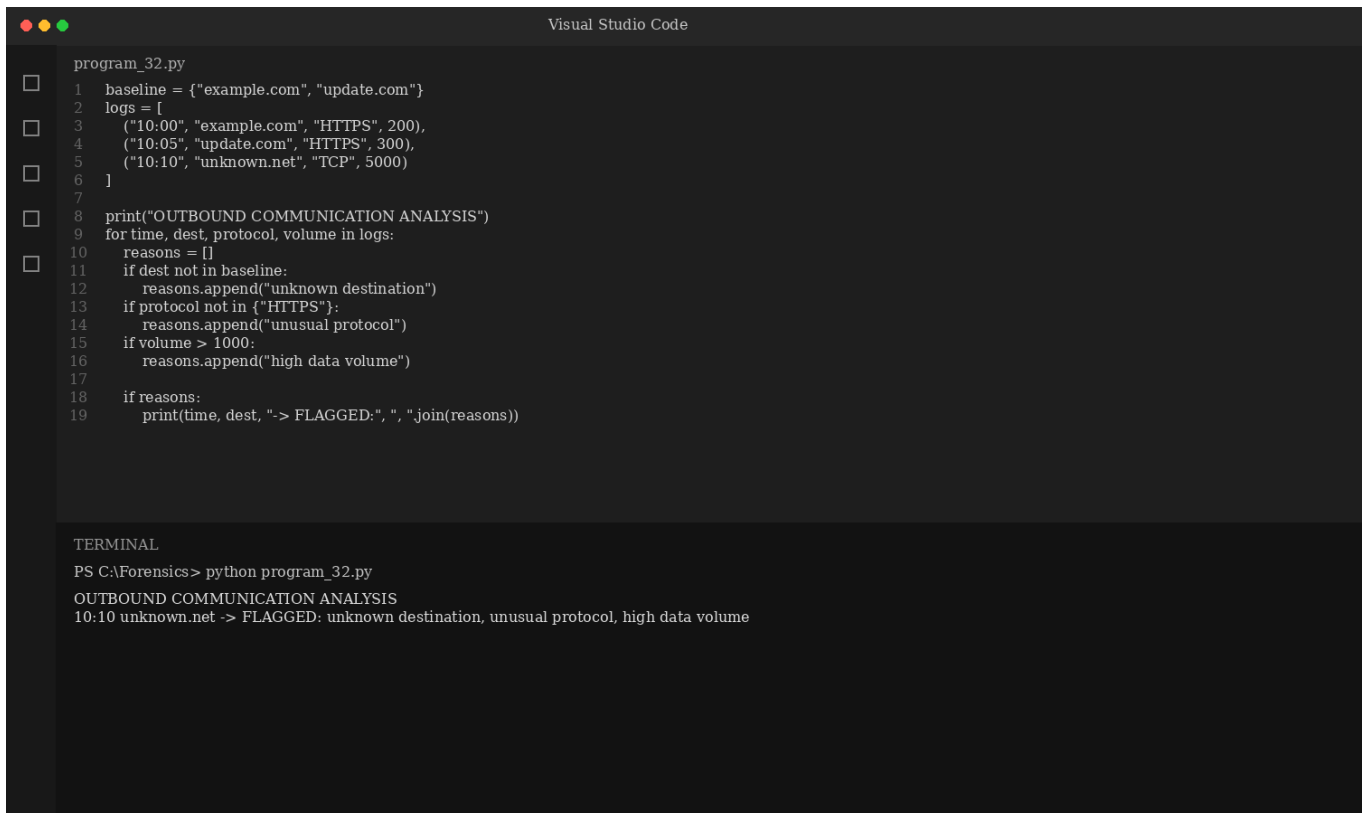
CODE

```
baseline = {"example.com", "update.com"}
logs = [
    ("10:00", "example.com", "HTTPS", 200),
    ("10:05", "update.com", "HTTPS", 300),
    ("10:10", "unknown.net", "TCP", 5000)
]

print("OUTBOUND COMMUNICATION ANALYSIS")
for time, dest, protocol, volume in logs:
    reasons = []
    if dest not in baseline:
        reasons.append("unknown destination")
    if protocol not in {"HTTPS"}:
        reasons.append("unusual protocol")
    if volume > 1000:
        reasons.append("high data volume")

    if reasons:
        print(time, dest, "-> FLAGGED:", ", ".join(reasons))
```

OUTPUT



The screenshot shows a Visual Studio Code editor window with a file named `program_32.py`. The code defines a baseline of domains and protocols, then iterates through a list of network logs. It checks each log entry against the baseline and flags any anomalies. The terminal output shows the program running and identifying the third log entry as flagged due to three reasons: unknown destination, unusual protocol, and high data volume.

```
program_32.py
1 baseline = {"example.com", "update.com"}
2 logs = [
3     ("10:00", "example.com", "HTTPS", 200),
4     ("10:05", "update.com", "HTTPS", 300),
5     ("10:10", "unknown.net", "TCP", 5000)
6 ]
7
8 print("OUTBOUND COMMUNICATION ANALYSIS")
9 for time, dest, protocol, volume in logs:
10     reasons = []
11     if dest not in baseline:
12         reasons.append("unknown destination")
13     if protocol not in {"HTTPS"}:
14         reasons.append("unusual protocol")
15     if volume > 1000:
16         reasons.append("high data volume")
17
18     if reasons:
19         print(time, dest, "-> FLAGGED:", ", ".join(reasons))
```

TERMINAL

```
PS C:\Forensics> python program_32.py
OUTBOUND COMMUNICATION ANALYSIS
10:10 unknown.net -> FLAGGED: unknown destination, unusual protocol, high data volume
```

RESULT

The program flags outbound communication that differs from the defined normal baseline.